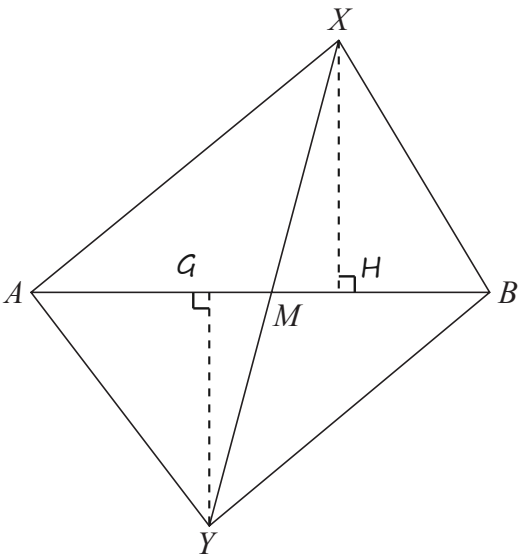
22 The diagram shows a quadrilateral $XBYA$.



The diagonals AB and XY intersect at the point M.

Given that the area of triangle AXB is equal to the area of triangle AYB, prove that XY is bisected by AB.

Draw XH and YG perpendicular to AB.

Equal areas and the common base AB give

½ × AB × XH = ½ × AB × YG, so XH = YG.

1 mark
Equal perpendicular heights

Angles XHM and YGM are both 90°.

Angles XMH and YMG are equal (vertically opposite).

1 mark
Equal angle pairs

Therefore triangles XHM and YGM are congruent (AAS).

1 mark
AAS congruence

Their corresponding hypotenuses are equal: XM = YM.

Hence AB bisects XY.

1 mark
XM=YM; bisected

(Total for Question 22 is 4 marks)